

SWERI & FAO COLLABORATION

Validation of WaPOR Evapotranspiration Data

in Northern Egypt — A Ground-Truthing Study

Dr. Tamer Hassan Khalifa

Senior Researcher, Soils, Water & Environment Res. Inst. (SWERI), Egypt

FAO WaPOR Phase 2 Project — "Monitoring Land and Water Productivity by Remote Sensing"

tamerkhalifa1985@gmail.com

2020 — 2024 Study Period



Background

Context

Egypt's Sustainable Agricultural Development Strategy (SADS) to 2030

2030 targets:

- Improve water-use efficiency in agriculture
- Reclaim new lands while protecting existing farmland
- Achieve near self-sufficiency in wheat, maize, sugar

The Gap

FAO's WaPOR V3 provides open-access, high-resolution (20m), near real-time data for monitoring water productivity across Africa and the Near East, but local validation is essential for building trust and enabling evidence-based agricultural planning and water resource management.

Agricultural Challenges



Water Scarcity

Limited freshwater resources



Climate Stress

Rising temperatures & aridity



Soil Degradation

Salinization & productivity loss



Data Trust Gap

Unvalidated RS ET estimates

SWERI + FAO Collaboration

Local validation of WaPOR V3 data using in-situ measurements to build trust for evidence-based water productivity planning in Egyptian agriculture.

Main Goal & Objectives

Primary Goal: Validate and verify WaPOR V3 evapotranspiration data using in-situ ground measurements across multiple validation sites in Northern Egypt.

1

Validate ETo and ETa Against Ground Stations

Compare WaPOR-derived evaporation (E) and actual evapotranspiration (ETa) with measurements from Cordoba meteorological station and Sakha energy balance instruments.

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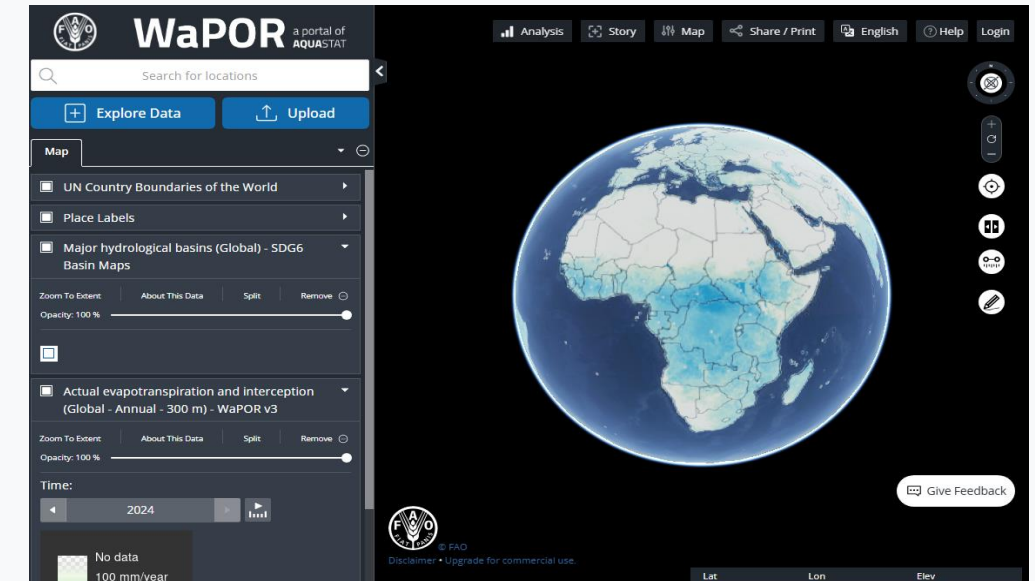
Multi-Dimensional Data Comparison

Cross-validate WaPOR outputs with auxiliary datasets: water balance (PCP vs AETI), Net Primary Productivity (NPP) yield estimates, and crop coefficient (Kc) analysis.

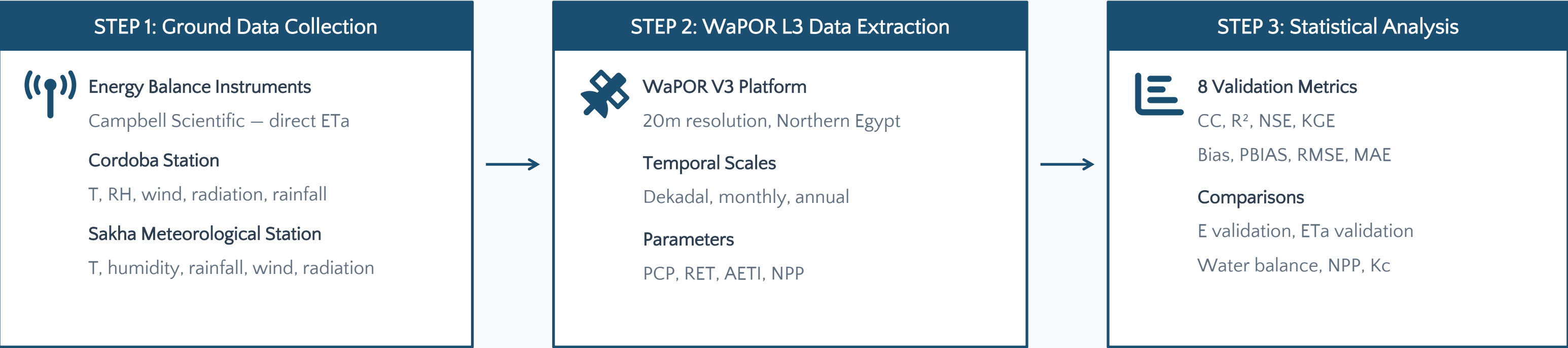
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Build Local Capacity

Train researchers on remote sensing science and WaPOR database technology; enable independent data processing and replication for tailored local solutions.



Methodology — Validation Workflow



Validation Metrics Used

Metric	Abbreviation	Ideal Value	Metric	Abbreviation
Pearson Correlation Coefficient	CC	1.0	Bias	—
Coefficient of Determination	R ²	1.0	Percent Bias	PBIAS
Nash-Sutcliffe Efficiency	NSE	1.0	Root Mean Square Error	RMSE

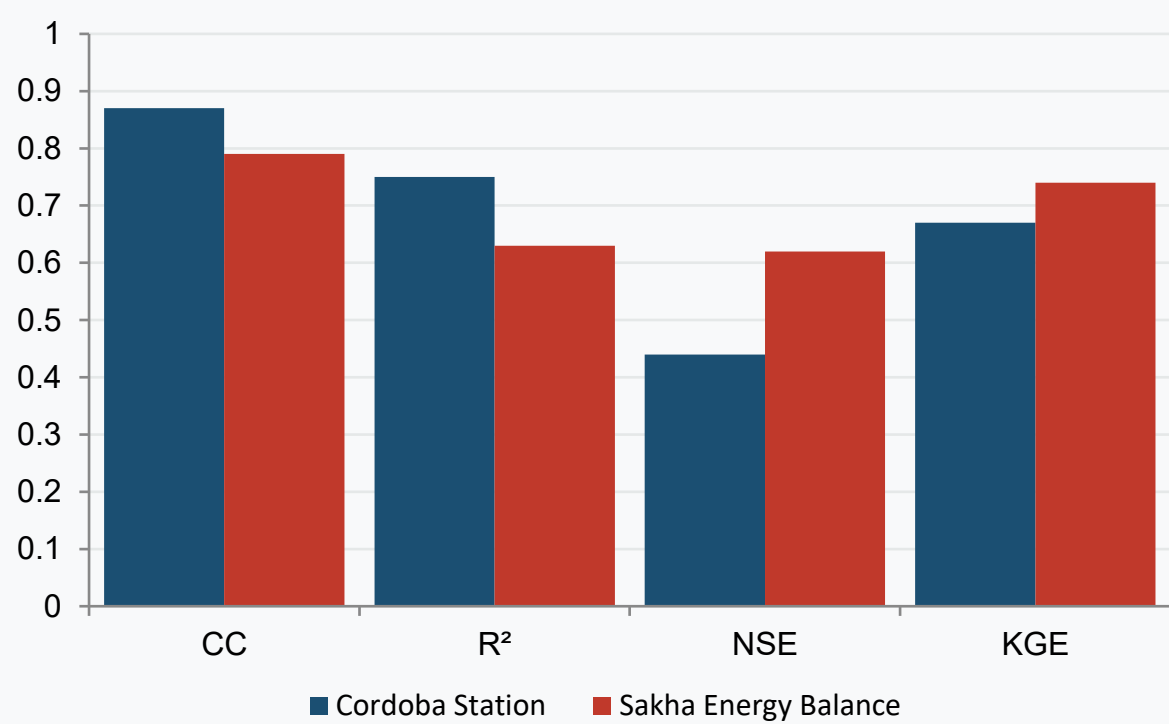
Kling-Gupta Efficiency (KGE) — Ideal: 1.0 | Mean Absolute Error (MAE) — Ideal: 0 | All metrics computed per standard hydrological validation protocols.

Results — Evaporation (E) Validation

Table 1. E Validation — Cordoba vs Sakha Station Metrics

Metric	Cordoba Station	Energy Balance
Data Points (n)	16	50
Average (mm)	23.5	2.6
Pearson CC	0.87	0.79
R ²	0.75	0.63
NSE	0.44	0.62
KGE	0.67	0.74
Bias	5.44	-0.08
PBIAS (%)	23.1	-3.1
RMSE	9.0	0.78
MAE	7.28	0.57

Fig. 1. Key Metrics Comparison



Key Findings

- ▶ Sakha energy balance shows superior NSE (0.62 vs 0.44) and KGE (0.74 vs 0.67)
- ▶ Cordoba shows higher R² (0.75) but large PBIAS (23.1%)
- ▶ Sakha has minimal bias (-0.08) and PBIAS (-3.1%)

Insight: Energy balance instruments (Sakha) provide more accurate validation reference than meteorological station data (Cordoba), with 10x lower PBIAS and 11x lower RMSE. Both stations show acceptable correlation (CC > 0.79).

Results — ETa Validation (Wheat 2020-2022)

Table 2. ETa Validation — Three Wheat Growing Seasons

Metric	Wheat 2020	Wheat 2021	Wheat 2022
Data Points (n)	16	15	15
Pearson CC	0.82	0.90	0.94
R ²	0.67	0.81	0.88
NSE	0.25	0.41	0.55
KGE	0.54	0.57	0.52
Bias	2.1	-2.97	2.51
RMSE	7.27	6.0	7.24
MAE	5.72	5.3	5.6

Fig. 2. Accuracy Trends Over Seasons

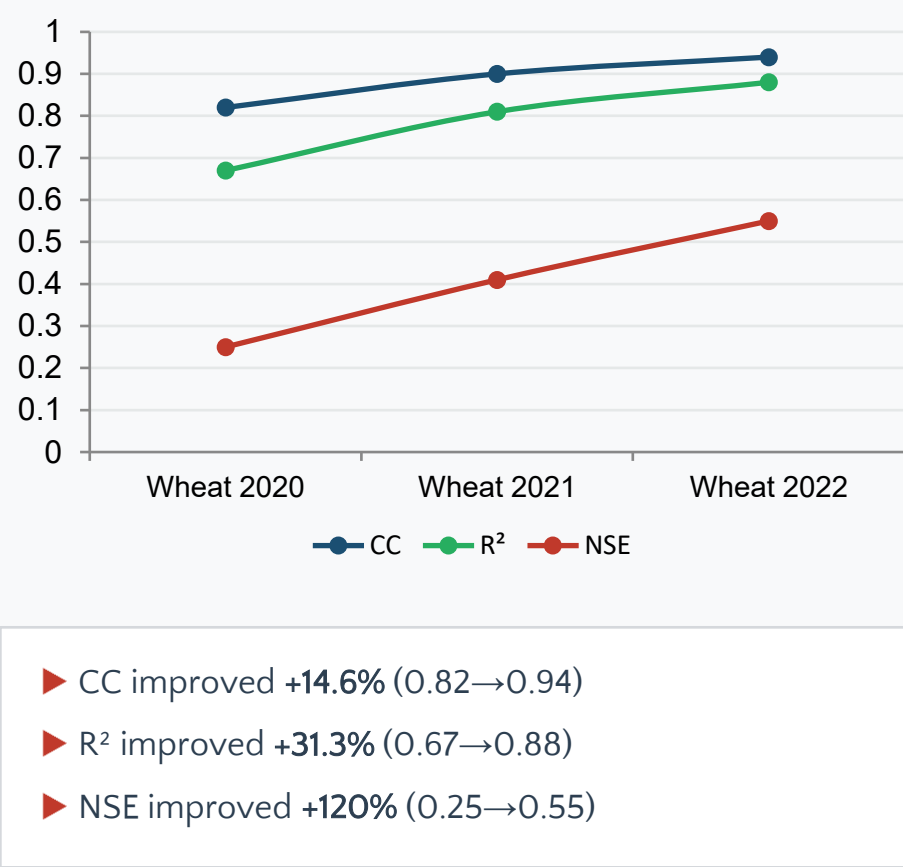
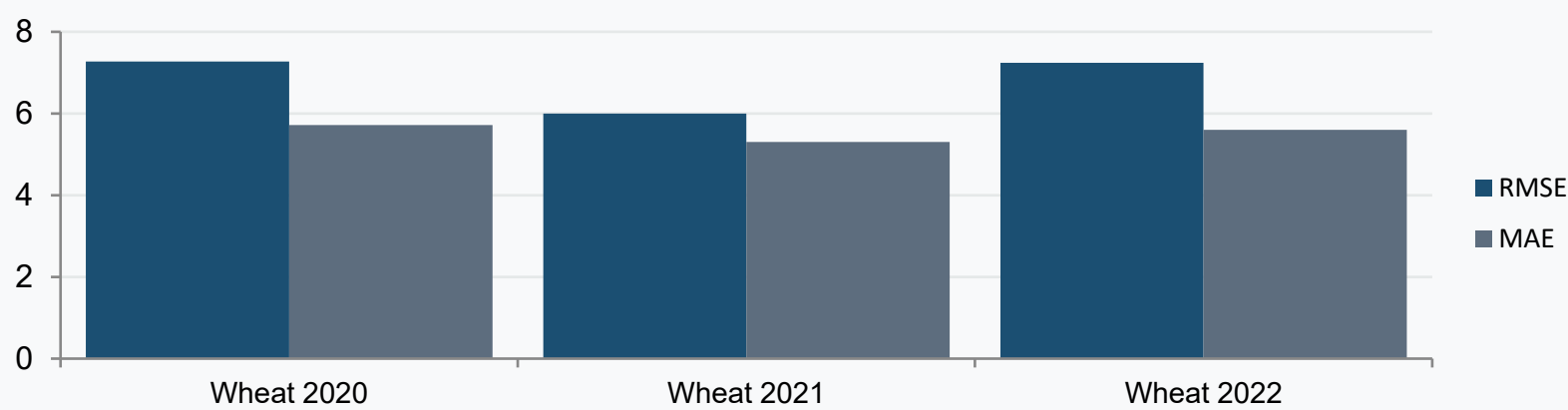


Fig. 3. Error Metrics Comparison (RMSE vs MAE)



KGE Stability: Kling-Gupta Efficiency remained stable across all three seasons (0.52–0.57), indicating consistent overall model performance despite varying error magnitudes.

Discussion — Key Trends & Interpretation

Trend 1: Improving Correlation Over Time

ETa correlation coefficient improved **15%** from 2020 (CC=0.82) to 2022 (CC=0.94). R² increased **31%** across seasons (0.67→0.88). This suggests either algorithm refinement in WaPOR V3 processing or more favorable climatic conditions during 2022.

Trend 2: Moderate NSE Indicates Systematic Bias

NSE remained moderate (0.25–0.55) despite high correlation, suggesting **systematic bias during peak ET periods**. WaPOR may under/over-estimate ETa during high-stress growth stages. NSE < 0.5 in 2020 and 2021 indicates suboptimal model efficiency.

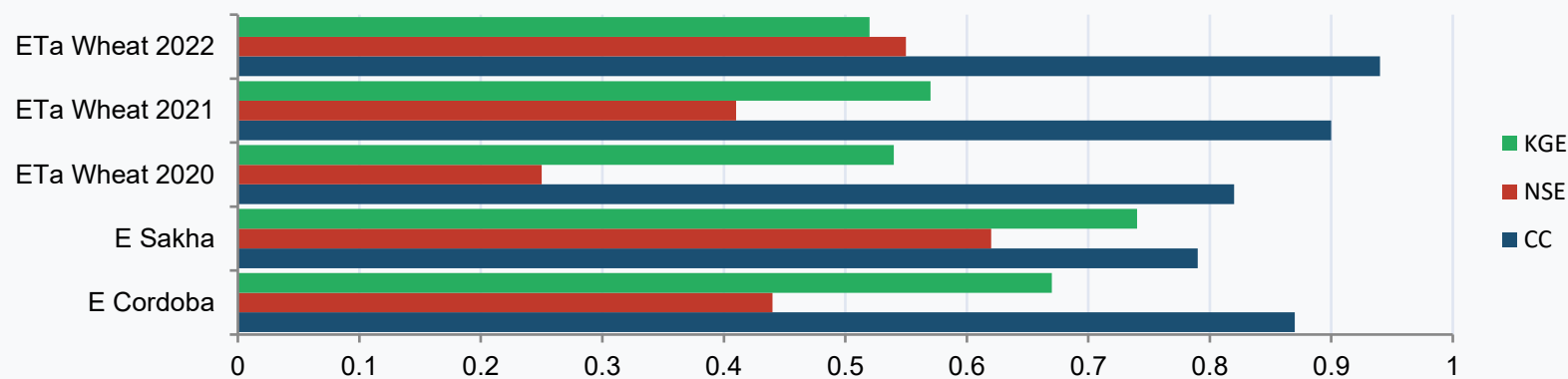
Trend 3: Energy Balance Outperforms Station Data

Sakha energy balance validation showed superior agreement (PBIAS=–3.1%, KGE=0.74) compared to Cordoba meteorological station (PBIAS=23.1%, KGE=0.67). Direct energy flux measurement provides more reliable ETa reference than derived meteorological estimates.

Trend 4: KGE Stability Across Conditions

KGE remained stable at 0.52–0.74 range across all validations, indicating **consistent model behavior**. The RMSE/MAE ratio (–1.25) suggests consistent proportional error magnitude, pointing to systematic rather than random deviation patterns.

Fig. 4. Metric Performance Summary — All Validation Datasets



Key Interpretation

WaPOR V3 demonstrates **strong and improving correlation** with ground ETa data (CC: 0.82–0.94). However, moderate NSE values reveal **systematic bias during peak ET periods** that should be addressed through local calibration.

Energy balance instruments provide the most reliable validation reference for future studies.

Conclusion — Quantified Takeaways

- 1** Strong correlation validated. WaPOR V3 ETa data shows strong correlation with ground measurements across all validation sites, with CC ranging from **0.82 to 0.94**.
- 2** Accuracy improved across seasons. Three-season wheat validation shows consistent improvement — R^2 gained **+31%** (0.67→0.88), NSE improved **+120%** (0.25→0.55).
- 3** Energy balance is the gold standard. Sakha energy balance validation outperformed meteorological station data with KGE of **0.74 vs 0.67** and PBIAS of **-3.1% vs 23.1%**.
- 4** NSE indicates refinement opportunity. Moderate NSE values (0.25–0.55) suggest room for **algorithm refinement** in peak ET period estimation, particularly for water-stress growth stages.
- 5** Local validation builds trust for policy. This study provides evidence-based confidence in WaPOR V3 for Egyptian agricultural water productivity planning and **SADS 2030 implementation**.

WaPOR V3 is validated for Egyptian agriculture — suitable for operational water productivity monitoring with noted calibration needs.

Limitations & Future Work

LIMITATIONS

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Crop-specific limitation
Validation limited to wheat crop across 3 growing seasons only
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Spatial coverage
Restricted to Northern Egypt; Nile Delta and desert fringes not covered
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Model bias in peak ET
NSE values (0.25–0.55) suggest systematic bias during high-ET periods
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Sample size variability
Data points range from 15–50 across validations; statistical power varies

FUTURE WORK

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Extend to other crops
Validate for rice, maize, sugarcane, and horticultural crops
- 

Multi-year climate analysis
Long-term validation across climate variability and El Niño cycles
- 

Eddy covariance integration
Incorporate eddy covariance towers for direct ET measurement validation
- 

Crop-specific Kc calibration
Develop locally-calibrated crop coefficients for Egyptian conditions

Research Impact Pathway

This validation study establishes the foundation for **operational use of WaPOR V3 in Egyptian agricultural planning**. Future work will expand the validation network, incorporate advanced measurement techniques, and develop crop-specific calibration protocols to maximize water productivity under the SADS 2030 framework.

Visual Summary — Key Findings at a Glance

0.79–0.94

Correlation Range (CC)

0.63–0.88

R² Range

5

Validation Datasets

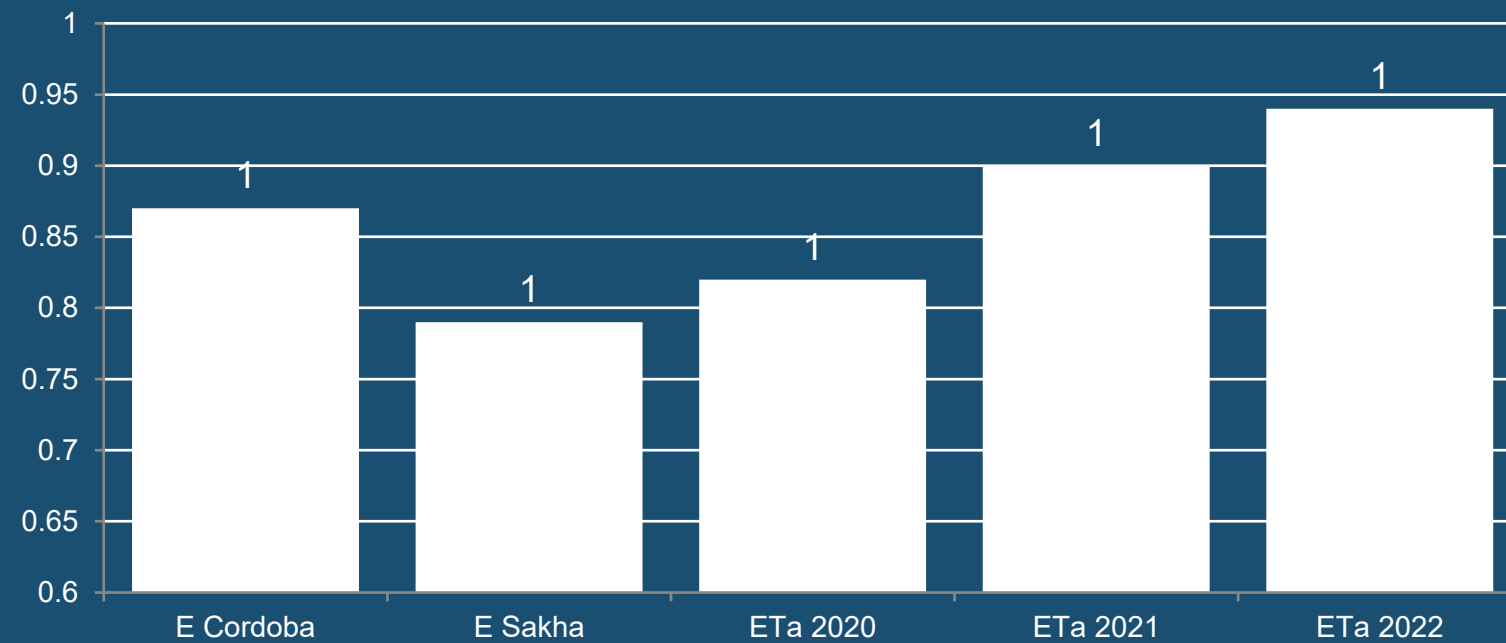
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Wheat Seasons

2

Ground Stations

Correlation Coefficient Across All Validations



Core Message

- ✓ WaPOR V3 validated for Egyptian agriculture
- ✓ Strong correlation: CC = 0.82–0.94
- ✓ Improving accuracy: R² +31% over 3 seasons
- ✓ Energy balance = best validation reference
- ✓ Ready for SADS 2030 implementation
- ✓ Local calibration needed for peak ET periods

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Thank You